

SECTION 8

RAMPING MAGNET TO FIELD



MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

8-1 INTRODUCTION

The 3.0T Corona magnet is ramped to a nominal current of 356 amperes to achieve a 3.0 Tesla field, using voltage control for ramping. The field parking range is recorded in the ATR (Acceptance Test Report), but will be 3.0000 ± 0.0005 Tesla ($30,000 \pm 5.0$ Gauss; $127.728 \text{ MHz} \pm 21.328 \text{ KHz}$).

The magnet is initially ramped until the helium level is reduced to 80%, then it is parked and refilled with liquid helium before finally being ramped to 3.0 Tesla. Make sure the magnet's helium level is 100% before beginning this procedure. Also make sure to plan for the helium fill while the magnet is parked before ramping to 3.0 Tesla. The parked fill will use approximately 500 liters of liquid helium and take about one hour.

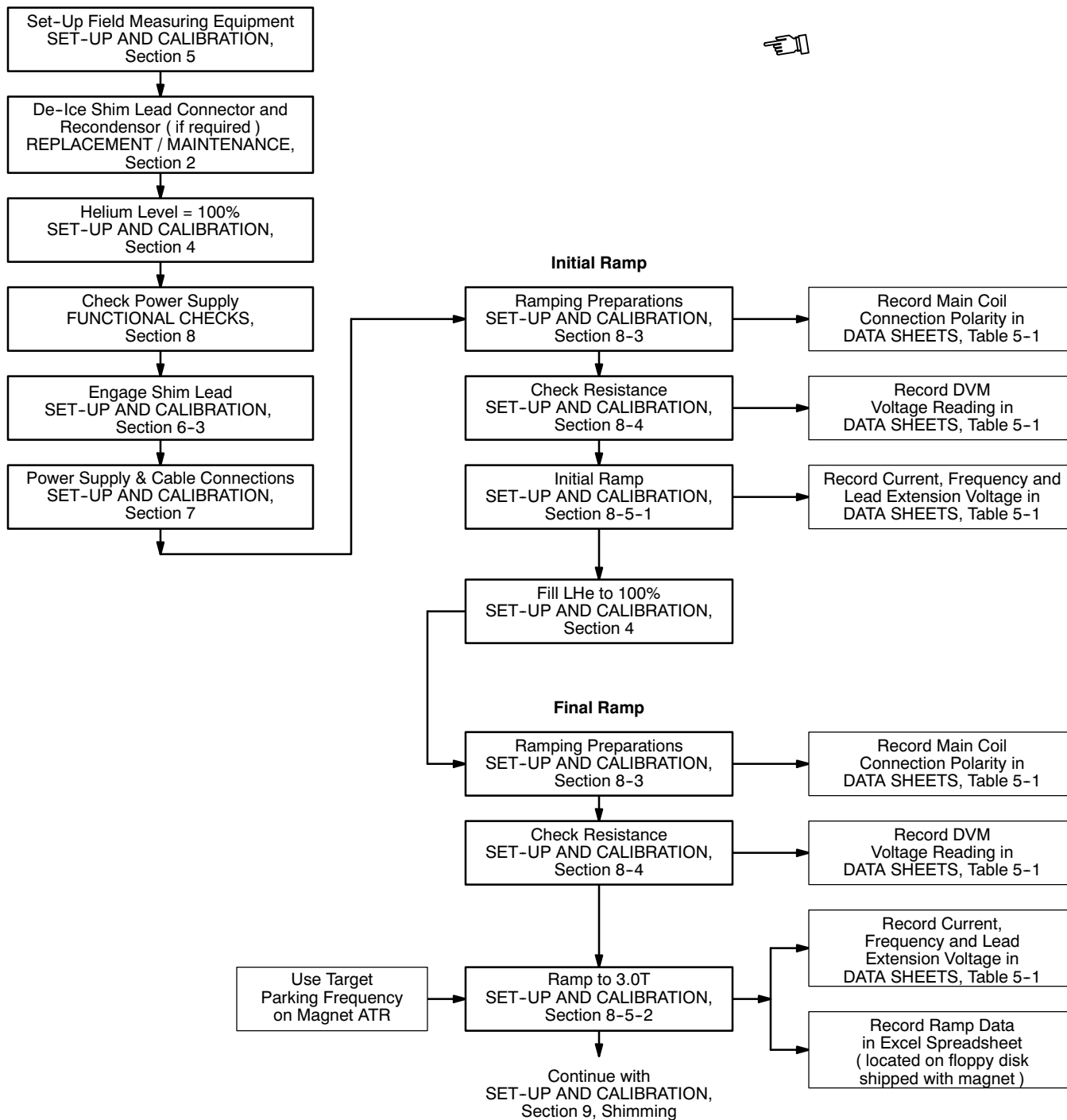
Initial ramp to 80% helium level will take approximately 6 hours, including all checks. The initial parking field will be 2.7 ± 0.1 Tesla ($27,000 \pm 1000$ Gauss; $114.96 \pm 4.26 \text{ MHz}$). Final ramp to 3.0 Tesla will take approximately 2 - 3 hours. A flowchart of ramping is shown in Illustration 8-1.

8-1 INTRODUCTION (continued)

WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING THE MAGNET:

- MAKE SURE THE MAGNET ROOM EXHAUST VENT FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM.
- NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.
- POST WARNING SIGNS OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, INCLUDED WITH THIS MANUAL.
- POST “RAMPED MAGNET” AND “AUTHORIZED PERSONNEL ONLY” WARNING SIGNS AT MAGNET ROOM ENTRANCE AND AT THE MR SUITE ENTRANCE, RESPECTIVELY, PRIOR TO RAMPING THE MAGNET. WARNINGS ARE TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED.
- REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW.
- MAKE SURE THE MAGNET RUNDOWN UNIT IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET-UP AND CALIBRATION, SECTION 1-5.
- MAKE SURE THE MAGNET IS 100% FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING TO A POINT, DURING RAMPING, WHERE A QUENCH MAY OCCUR.
- REVIEW AND FOLLOW SAFETY MEASURES CONTAINED IN 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, INCLUDED WITH THIS MANUAL.

8-1 INTRODUCTION (continued)

RAMPING PROCESS FLOWCHART, BEFORE RAMPING
ILLUSTRATION 8-1

8-2 TOOLS / EQUIPMENT

8-2-1 Safety Equipment

- 3.0T Warning Sign & Label Kit 2379494
- Portable Oxygen Monitor 2287000
- Cryogen Safety Wear Kit 46-271137G1
- Nonferrous Shoes

8-2-2 Other Tools / Equipment

- Main Coil Power Supply:
 - 750 Amp Model TCR 7.5T750 (46-260776G3) or
 - 1000 Amp Model ESS 7.5-1000-2-D-1236 (46-260776G4)
- 750 Amp Ramp Cart / Cable Kit 2135435
- Shim Case / Cable Kit 2135558
- Magnet Ramping Equipment Kit 46-260703G5 (for standard ceiling height sites), which includes:
 - Fixed Main Lead Extension 46-294204G1
 - Hold-Down Tool 2193922
 - Retractable Shim Lead Assembly 2285073
- 2.5M Ceiling Height Kit M1060SR (if required for low ceiling height sites), which includes:
 - Flexible Main Lead Extension 2287029
 - Hold-Down Tool, Short, 2276755
 - Retractable Shim Lead Assembly, Short, 2285073
- LCC300 Ramp Lead Hold-Down Tool 5116737 (only for use when LHe topoff is planned immediately after parking the magnet in Section 8-5-1)
- Digital Voltmeter (DVM)
- MetroLab Teslameter Kit 46-251865G5 or equivalent magnetic field measuring device.
- #6 MetroLab Field Probe 2295525-5 (included in 46-251865G5)
- #5 MetroLab Probe 46-255839P104 (included in 46-251865G5)
- LCC300 Field Mapping Fixture 2380758 (included in LCC300 Passive Shimming Kit 2385710)
- Nonmagnetic Tools Kit 46-320273G4

8-3 RAMPING PREPARATIONS

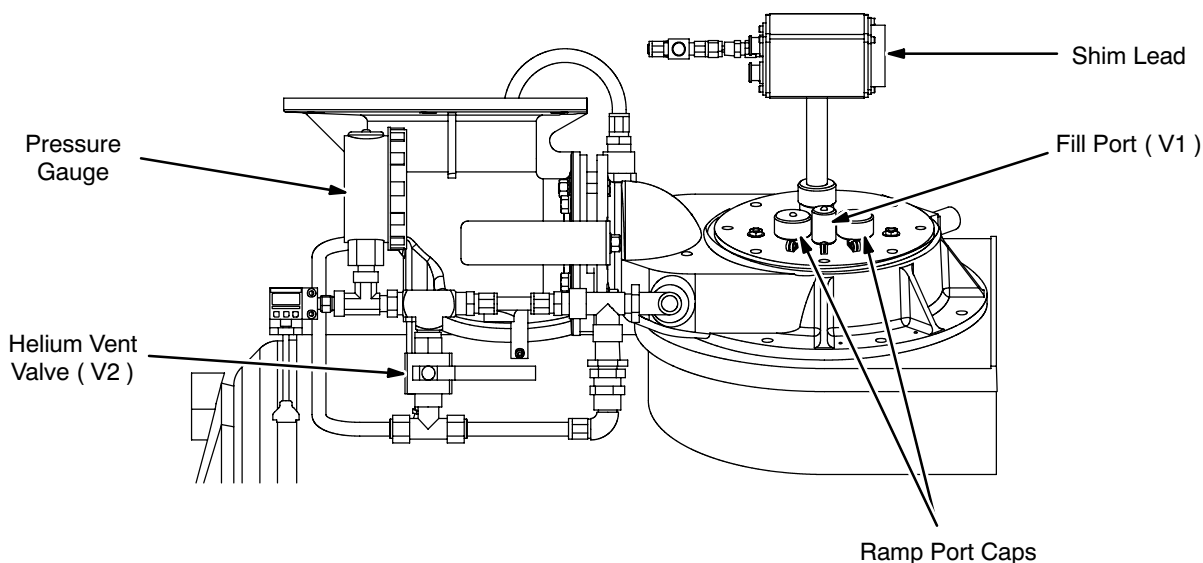
IMPORTANT !!!

Make sure the area is secure and that ALL required Warning Signs are posted before starting this procedure.

IMPORTANT !!!

Before ramping:

- Perform Shim Lead connector checks in conformance with FUNCTIONAL CHECKS, Section 5, Magnet Electrical Checks. If an ice blockage is found, deice in conformance with REPLACEMENT / MAINTENANCE, Section 2-3, Shim Lead Connector and Vertical Penetration Deicing.
 - Purge the Recondenser with helium gas in conformance with REPLACEMENT / MAINTENANCE, Section 2-4-5, Level Three Deicing.
1. Make sure helium level readings on Magnet Monitor are functional and accurate. If problems are found, troubleshoot before ramping.
 2. Set up the Metrolab Teslameter and Probe in conformance with SET-UP AND CALIBRATION, Section 5.
 3. Check for ice on the Shim Lead connector and the Recondensor. Deice in conformance with REPLACEMENT / MAINTENANCE, Section 2, Ice Removal, if required, before ramping the magnet.
 4. Make sure the magnet is 100% full of helium and Vent Valve V2 is closed before ramping the magnet. See Illustration 8-2.



VALVE / VENT LOCATIONS
ILLUSTRATION 8-2

8-3 RAMPING PREPARATIONS (continued)**WARNING!**

THE MAGNET CAN BE QUENCHED IF THE MAGNET POWER SUPPLY EXPERIENCES LARGE OUTPUT VOLTAGE FLUCTUATIONS AND / OR EXCESSIVE RIPPLE. MAKE SURE THE POWER SUPPLY IS ROUTINELY CALIBRATED AT AN APPROVED FACILITY.

5. Make sure the Main Power Supply is installed, checked and adjusted in conformance with the vendor manual supplied with the unit and with FUNCTIONAL CHECKS, Section 8, Main Power Supply Test.
6. Make sure the Shim Lead Assembly is “Engaged” in conformance with SET-UP AND CALIBRATION, Section 6-3, Shim Lead Engagement.
7. Make sure the Input Power Cable to the Main Power Supply is disconnected.

WARNING!

A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.

MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:

- **SLOWLY INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE CONTACTING THE MAIN LEAD PINS.**
 - **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
 - **WEAR NON-ABSORBANT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
 - **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
8. Connect the power supply to the magnet by making all cable connections in conformance with SET-UP AND CALIBRATION, Section 7-4, Main Coil Power Supply Connections to Magnet.

8-3 RAMPING PREPARATIONS (continued)

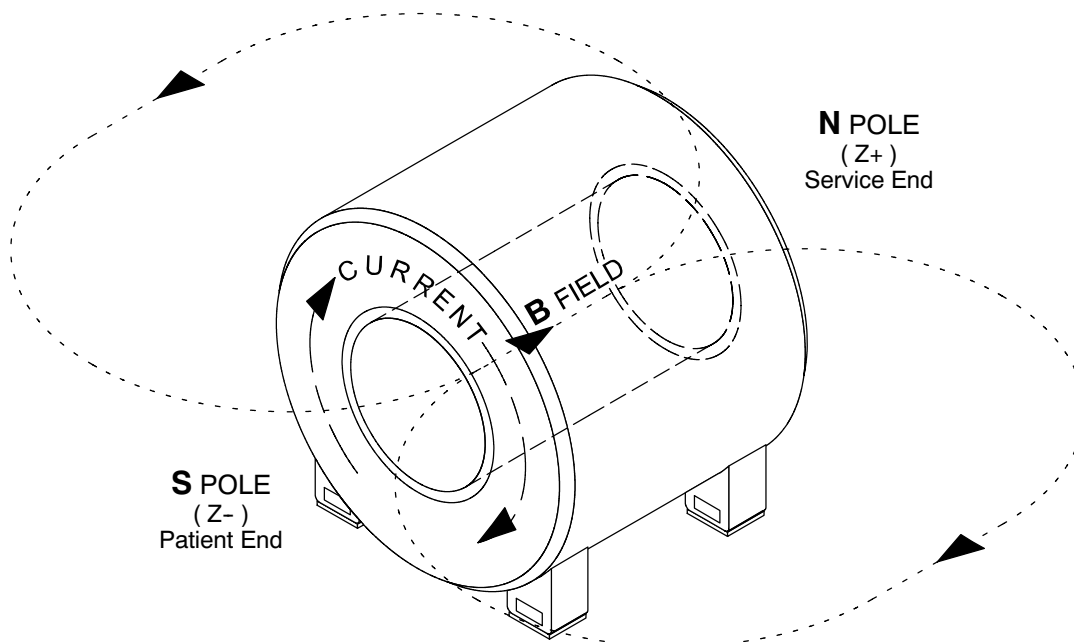
9. Record the Main Coil connection polarity in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.
10. Set all heater switches on the power supply to the "OFF" position. Set the Current Adjust and Voltage controls to zero (full CCW).
11. Connect the Input Power Cable to the Main Power Supply.

8-4 RESISTANCE CHECKS

1. If the magnet is not at field, check the Switch Heater and Shim Coil resistances in conformance with FUNCTIONAL CHECKS, Section 5, Magnet Electrical Checks.
2. Make sure the Current Adjust and Voltage controls on the Main Power Supply are "OFF" (full CCW).

Note

All main coil voltages provided in this procedure will be equal in magnitude but opposite in polarity for "reverse ramped" magnets. Normal polarity is shown in Illustration 8-3.



NORMAL RAMPED MAGNET POLARITY
ILLUSTRATION 8-3

8-4 RESISTANCE CHECKS (continued)**WARNING!**

MAKE SURE THE MAIN HEATER SWITCH IS OFF DURING RESISTANCE CHECKS.

3. Set the Main Power and Power On switches to "ON."

Note

There are duplicate Main Power and Power On switches on the front and back of the Main Power Supply.

4. Make sure the Main Heater Switch is off, then set the Heater 2 Shim Axial Switch to "I" (on) and observe the current rise in the Main Power Supply Ammeter (700 - 720 mA) to verify circuit continuity.
5. Connect a Digital Voltmeter (DVM) to the end of the Voltage Sense Leads.
6. Set the Current Adjust Coarse Control on the power supply to maximum (full CW).
7. Observe the Main Power Supply Ammeter and slowly turn the Voltage Control clockwise to set a 375A current through the Main Power Leads, Main Lead Extensions and persistent Main Switch.
8. Record the voltage reading on the DVM in the DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.

8-4 RESISTANCE CHECKS (continued)**WARNING!**

A VOLTAGE READING GREATER THAN 100 MILLIVOLTS AT 375 AMPS INDICATES UNACCEPTABLE INTERNAL CONTACT RESISTANCE OF THE LEAD EXTENSIONS. HIGHER RESISTANCES WILL ADD MORE HEAT TO THE MAGNET, INCREASING BOILOFF AND POSSIBLY CAUSING A QUENCH DURING RAMPING.

9. Perform one or more of the following steps as necessary if the DVM voltage is greater than 100mV.
 - Wait approximately 1 minute with current running, the DVM readings may drop as the Power Lead Extensions cool.
 - If the reading still exceeds 100 mV:
 - a. Turn the Voltage and Current Adjust controls to zero (full CCW).
 - b. Turn off the Magnet Power Supply input power.
 - c. Tighten the bolts securing the Ramp Cables to the Power Supply and Main Leads Extensions.
 - d. Lift and reseal the Main Leads.
 - Repeatedly failing the contact resistance check may indicate one of the following:
 - a. The Main Lead Extension Contact Bands replaced as directed in SET-UP AND CALIBRATION, Section 7-4, are not performing correctly and need replacing in conformance with REPLACEMENT / MAINTENANCE, Section 9.
 - b. The Main Leads are damaged and need replacing.
 - Repeat Steps 6 through 9.
10. Continue to Step 11 after the DVM voltage is < 100mV.
11. Set the power supply's Voltmeter Select Switch to "MAIN POWER SUPPLY" position. This will display the output at the output lugs of the power supply monitored. A voltage < 2.2V at 375A indicates acceptable system resistance. If the voltage exceeds 2.2V during the test, follow the procedures in Step 9 for adjusting contact resistance.

8-4 RESISTANCE CHECKS (continued)

12. Adjust the Voltage Adjust Control to pass 375A through the Main Power Leads, Main Lead Extensions and persistent Main Switch while observing the power supply's Voltmeter. If the voltage exceeds 2.2V, then retighten the bolts securing the Ramp Cables to the Power Supply and Main Lead Extensions.
13. Turn the Current Adjust and Voltage Adjust controls off (full CCW).

8-5 RAMPING



If a quench occurs during ramping, immediately:

- a. Turn the Main Coil Power Supply Voltage and Current Controls to zero (fully CCW),
- b. Turn off the Power Supply,
- c. Leave the room and
- d. Turn off the Cryocooler Compressor outside the magnet room.

Do not reenter the magnet room until the quench is over. A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the Vent System.



MAKE SURE THE AXIAL SHIM SWITCH HEATERS ARE "ON" DURING RAMPING TO PREVENT COIL DAMAGE AND A MAGNET QUENCH. GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE "OFF" POSITION. THE SHIM POWER SUPPLY IS REQUIRED FOR TRANSVERSE HEATERS.

Note

Ice will form around the flow holes in the Ramp Lead Hold-Down Tool during ramping. Remove ice as needed to maintain helium gas flow.

8-5-1 Initial Ramp & Fill

1. Make sure the power supply controls are set as follows:
 - a. Voltage Control zero (full CCW)
 - b. Current Control 400 amps
 - c. Main Power Switches on the Main Ramp Power Supply "ON"
 - d. Heater 2 Shim Axial Switch on the Main Power Supply "I" (on)
 - e. Voltmeter Select Switch on the power supply "Main Coil" position
2. Set the Heater 1 Main Switch to "I" (on). Wait 3 minutes.
3. Use a Range 5 Teslameter Probe (46-255839P104) to monitor field strength below 2.0 Tesla, then use a Range 6 Teslameter Probe (2295525-5) to monitor field strength above 2.0 Tesla.
4. Ramp the magnet in conformance with Table 8-1 as follows:
 - a. Slowly turn the Voltage Control to obtain the Main Coil (Ramp) Voltage value shown in the table, starting with 6.00 Volts. Allow the voltage to decay over the current interval shown in the table.
 - b. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Control to obtain the Main Coil Voltage for the next current range.
5. When a ramp current of 50 amps is reached, estimate the system inductance by measuring the ramp current change over a 60 second interval.

$$L \text{ (Inductance) } = 60 \times \text{Voltage} / \text{Current Change}$$

If the measured inductance (L) is outside the range of 200 – 230 Henries, contact your regional MAC Team representative.

TABLE 8-1
RAMPING PROFILE: VOLTAGE VERSUS CURRENT

Main Coil (Ramp) Voltage (Volts)	Main Coil (Ramp) Current Range (Amps)
6.00	0 – 248
4.00	248 – 283
2.00	283 – 314
1.00	314 – 342
0.50	342 – 360
0.30	> 360*
* When 2.9995 Tesla is reached, lower Main Coil voltage to maintain 2.9995 – 3.0005 Tesla field.	

8-5 RAMPING (continued)**8-5-1 Initial Ramp & Fill (continued)****WARNING!**

THE 3T MAGNET CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING. ADJUST THE VOLTAGE CONTROL IN A SLOW, CONTINUOUS MOTION IN THE FOLLOWING STEP.

6. Closely monitor the liquid helium level in the magnet during ramp. When the level is reduced to 80%, immediately stabilize the field (which should be 2.6 – 3.0 Tesla), park the magnet and disconnect / remove ramping connections and apparatus as follows:
 - a. Slowly adjust the Voltage Adjust Control to stabilize the magnet field. The last two digits on the Teslameter should be the only digits changing.
 - b. Maintain the field at the stable setting for 5 minutes before proceeding to Step 6c.

Note

Observe the Power Supply Voltmeter reading with the Voltmeter Select Switch in the “Main Coil” position. When the switch goes ‘persistent’ the voltage across the magnet terminals will stabilize at 0.00.

- c. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go ‘persistent.’

WARNING!

MAKE SURE THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN DATA SHEETS, TABLE 5-1, MAGNET RAMPING & PARKING CURRENT LOG. THIS INFORMATION IS ESSENTIAL FOR LATER CHANGING THE MAGNETIC FIELD. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH. (NORMAL POLARITY IS SHOWN IN ILLUSTRATION 8-3.)

- d. Record the current, frequency and lead extension voltage values at which the switch went ‘persistent’ in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.

8-5 RAMPING (continued)**8-5-1 Initial Ramp & Fill (continued)**

6. (continued)

- e. When the switch is “persistent”, **slowly** turn the power supply’s Voltage Adjust Control to zero (full CCW) over a two minute period.

Note

Check that the Teslameter reading does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

- f. Gradually turn the Current Adjust Control to zero (over a one minute period).
- g. Set the Heater 2 Shim Axial Switch to “O” (off).
- h. Set the Main Power and Power On switches on the Main Power Supply to “OFF” and disconnect the Input Power Cable.

IMPORTANT !!!

LCC300 Ramp Lead Hold-Down Tool 2343081 allows for immediate the helium fill while the Main Lead Extensions are in place. If this clamping fixture is being used, skip Steps 6i and 6j.

- i. Remove the Ramp Lead Hold-Down Tool (except if LCC300 Ramp Lead Hold-Down Tool 2343081 is being used).

8-5 RAMPING (continued)**8-5-1 Initial Ramp & Fill (continued)**

6. (continued)

WARNING!

A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.

MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:

- **SLOWLY INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE CONTACTING THE MAIN LEAD PINS.**
- **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
- **WEAR NON-ABSORBANT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
- **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**

j. Remove the Main Lead Extensions and cap the Ramp Lead Ports (except if LCC300 Ramp Lead Hold-Down Tool 5116737 is being used).

7. Fill the magnet to 100% in conformance with SET-UP AND CALIBRATION, Section 4, Liquid Helium Fill. The magnet will require approximately two 250 liter Dewars of liquid helium, and the fill will take approximately one hour.

8-5 RAMPING (continued)

8-5-2 Ramp to 3.0T

1. Make sure all steps (1 - 11) in Section 8-3, Preparation, have been completed.
2. Perform the resistance checks in Section 8-4.
3. Make sure the power supply controls are set as follows:
 - a. Voltage Control zero (full CCW)
 - b. Current Control 400 amps
 - c. Main Power Switches on the Main Ramp Power Supply "ON"
 - d. Heater 2 Shim Axial Switch on the Main Power Supply "I" (on)
 - e. Voltmeter Select Switch on the power supply "Main Coil" position
4. Use a Range 6 Teslameter Probe (2295525-5) to monitor field strength. The probe will lock in on the field around 1.0 Tesla (42.5764 MHz; 120 amps).
5. Retrieve the Main Coil connection polarity and latest parking current from DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.

Note

The Axial Shim Heater must remain on throughout the ramping procedure.

6. Make sure the Main Leads are connected with the polarity indicated in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.
7. Set the Current Adjust Controls on the Main Power Supply to 400 amps.
8. Set the Voltage Adjust Control to adjust the power supply output current to the parking current value obtained from DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.
9. Set the Heater 1 Main Switch to "I" (on).
10. Allow approximately 3 minutes for the Main Switch to go normal.

8-5 RAMPING (continued)**8-5-2 Ramp to 3.0T (continued)**

11. Ramp the magnet in conformance with Table 8-1 as follows:
 - a. Slowly turn the Voltage Adjust Control to obtain the Main Coil (Ramp) Voltage value shown in the table. Allow the voltage to decay over the current interval shown in the table.
 - b. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Adjust Control to obtain the Main Coil Voltage for the next current range.
12. When the magnet field reaches 2.9995 Tesla, slowly adjust the Voltage Adjust Control for 0.1 volts main coil driving voltage as indicated on the power supply's Voltmeter.
13. When the magnet field reaches 2.9995 – 3.0005 Tesla, stabilize the field and park the magnet:
 - a. Slowly adjust the Voltage Adjust Control for 0.025 volts main coil driving voltage as indicated on the power supply's Voltmeter. Adjust the voltage to maintain 2.9995 – 3.0005 Tesla.

Note

The optimum magnet parking field can be predicted from the correction coil currents used in Florence. Refer to the Acceptance Test Report (ATR) for the target parking frequency.

- b. Check the Teslameter and slowly adjust the Voltage Adjust Controls, as required, to bring the magnetic field to the Tesla value stated in the ATR. Allow the final field to stabilize; the last two digits on the Teslameter should be the only digits changing.
- c. Maintain the field at the final setting for 5 minutes before proceeding to Step 13d.

Note

Observe the voltage reading on the power supply's Voltmeter (with its toggle switch in the "Main Coil" position). When the switch goes 'persistent,' the voltage across the magnet terminals will stabilize at 0.00.

- d. Turn off the Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go 'persistent.'
- e. Record the current, frequency and lead extension voltage values at which the switch went 'persistent' in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Current Log.

8-5 RAMPING (continued)**8-5-2 Ramp to 3.0T (continued)**

13. (continued)

WARNING!

MAKE SURE THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN DATA SHEETS, TABLE 5-1, MAGNET RAMPING & PARKING CURRENT LOG. THIS INFORMATION IS ESSENTIAL FOR CHANGING THE MAGNETIC FIELD LATER. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH. (NORMAL POLARITY IS SHOWN IN ILLUSTRATION 8-3.)

Note

Check that the Teslameter reading does not decrease as the Voltage Adjust Control is turned to zero. Only the last two digits on the Teslameter should change. If the field decreases as the Voltage Control is turned, the Main Coil Switch is not persistent; slowly adjust the Voltage Control to return to parking field. The field will drop approximately 1 KHz when the power supply is being dialed down to zero amps.

- f. When the switch goes 'persistent,' slowly turn the power supply's Voltage Adjust Control to zero (full CCW) over a 2 minute period.
- g. Gradually turn the Current Adjust Control to zero over a one minute period.
- h. Set the Heater 2 Shim Axial Switch to "O" (off).
- i. Set the Main Power and Power On switches on the Main Power Supply to "OFF." Disconnect the Input Power Cable.
- j. Remove the Ramp Lead Hold-Down Tool.

8-5 RAMPING (continued)**8-5-2 Ramp to 3.0T (continued)****WARNING!**

A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.

MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:

- **SLOWLY INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE CONTACTING THE MAIN LEAD PINS.**
- **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
- **WEAR NON-ABSORBANT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
- **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**

CAUTION

Follow Step 14 precisely to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

14. After parking the magnet and disconnecting the main power supply, plug and remove one Main Extension at a time in the following sequence:
 - a. Open valve V2. Bleed vessel pressure down below 1.0 psig. Do not allow pressure to go below 0.5 psig as read on the pressure controller. Close V2.
 - b. Remove all ice around the ramp lead port compression fitting on the Main Lead Extension that is being removed (i.e., the Main Lead Extension that will have its flow hole plugged in Step 14c). See Illustration 8-4.
 - c. Install a screw into the flow hole of the Main Lead Extension deiced in Step 14b. See Illustration 8-4.
 - d. Unscrew the ramp lead port compression fitting and remove the Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.

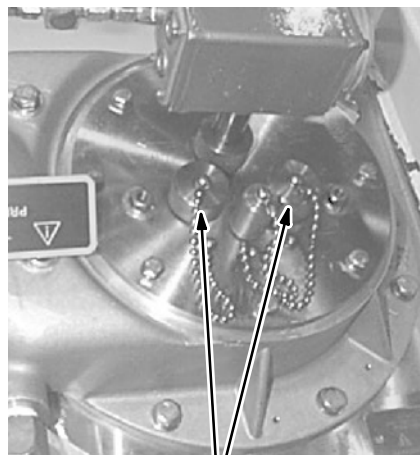
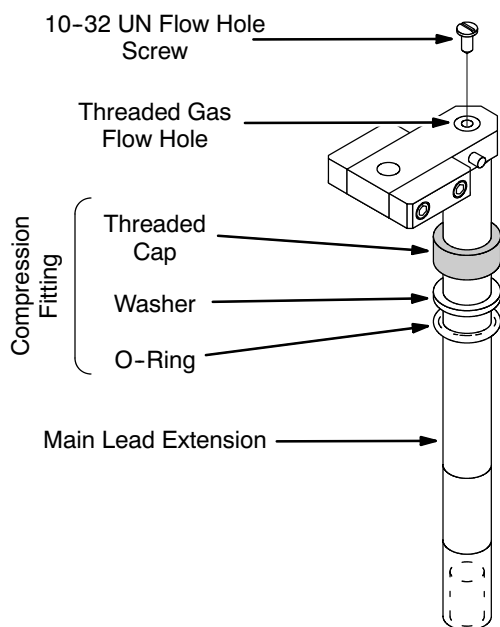
8-5 RAMPING (continued)**8-5-2 Ramp to 3.0T (continued)**

14. (continued)

- e. Repeat Steps 14b through 14d for the other Main Lead Extension.
- f. Check the Main Leads and Fill Port Cap for leaks.

CAUTION

Make sure any currents induced in the Transverse Coils during ramping are removed in conformance with SET-UP AND CALIBRATION, Section 9-3-3, Field Stability Check, before centering the Teslameter Probe and shimming the magnet.



Ramp Port Caps

MAIN LEAD EXTENSION AND RAMP PORT COMPRESSION NUT
ILLUSTRATION 8-4

8-5 RAMPING (continued)**8-5-2 Ramp to 3.0T (continued)**

Cryostat exhaust flow rate and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the helium exhaust system or Vent Valve V2.

15. Plug in the heater on the Magnet Monitor with a setting of 4.0 psig.

Note

If a shorter pressure rise time is needed, turn on the Axial Heater on the Main Power Supply OR, in the case of the Shim Power Supply, turn on both the Axial and the Transverse Heaters.

16. Continue with SET-UP AND CALIBRATION, Section 9, Shimming, setting up for and taking the first field measurement of the field stability check.

Note

If LCC300 Field Mapping Fixture 2380758 has not been installed prior to ramping, do so in conformance with SET-UP AND CALIBRATIONS, Section 5-4, Preparation for Field Mapping, before beginning the 12-hour post-ramp field stability check.

IMPORTANT !!!

The post-ramp field stability check performed in SET-UP AND CALIBRATION, Section 9-3-3, takes 12 hours. Set up and take the first measurement in that check as soon after ramping is finished as possible.